SUPPLEMENT TO "UNIFORM SADDLEPOINT EXPANSIONS FOR MULTIVARIATE TAIL PROBABILITIES"

BY SIMON A. BRODA[1,a] AND RAYMOND KAN[2,3,b]

[1]*Institute of Financial Services IFZ, Lucerne University of Applied Sciences,* [a]*simon.broda@hslu.ch*

[2]*Rotman School of Management, University of Toronto,* [b]*raymond.kan@rotman.utoronto.ca*

[3]*School of Business and Management, Hong Kong University of Science and Technology*

Section S1 gives the Lugannani–Rice expansion in the notation of the paper, with its proof, the series for its coefficients that shows the cancellation of their singularities at the mean, and the identity that relates its derivative to Daniels' expansion. Section S2 compares the univariate approximations of orders one to six with exact values on large grids for the noncentral chi-squared and the singly and doubly noncentral F distributions. Section S3 gives details on the bivariate examples of Wang (1990) and of Kolassa and Li (2010): the behavior of the expansions and of their approximations with the point fixed and the sample size doubling, the direct evaluation at a saddlepoint with a negative component, the singular lines of Kolassa and Li's formula, and the Laplace example. Section S4 applies the expansions to the sample autocorrelations at two lags, Section S5 gives the computational details, and Section S6 describes the programs.

S1. The Lugannani–Rice expansion. This section gives the Lugannani–Rice expansion of Remark 1 of the paper with its proofs, in the notation of Section 4 of the paper: $\hat{s}$ is the saddlepoint, $\hat{K}_k = K^{(k)}(\hat{s})$, θ_k, $\hat{w}$ and $\hat{u}$ are the standardized quantities (26), $\Theta(s) = \sum_{k=1}^{\infty} \theta_{k+2} s^k$, $d_{r,j}$ and $a_{i,j}$ are the coefficients (27) and (29), $h_i = \mathrm{He}_{2i}(0)$, $\hat{u}_n = \sqrt{n}\hat{u}$ and $\hat{w}_n = \sqrt{n}\hat{w}$, and $\mu = E(X)$. The univariate inversion integral (33) has a pole at $s = 0$, and the two classical expansions differ in how they treat it: the Hauschildt–Robinson expansion (Corollary 2 of the paper) keeps the pole and expands only the exponential factor, while the expansion of Lugannani and Rice (1980) removes it by subtracting a function with the same pole whose integral is a normal tail probability, and expands the remainder. Written with the $a_{i,j}$ it is the following.

THEOREM S1 (Lugannani–Rice expansion). *For $\bar{x} \neq \mu = E(X)$,*

$$\mathbb{P}(\bar{X} \leq \bar{x}) = \Phi(\hat{w}_n) - \phi(\hat{w}_n) \sum_{i=0}^{M-1} \frac{b_i}{n^{i+1/2}} + O(n^{-M}), \tag{S1}$$

where

$$b_i = \sum_{j=0}^{2i} \frac{(-1)^j a_{i,j}}{\hat{u}^{2i+1-j}} - \frac{h_i}{\hat{w}^{2i+1}},$$

and the $O(n^{-M})$ is relative to the leading term and uniform for $\hat{s}$ in compact subsets of the domain of K that do not contain 0. Moreover

$$b_i = \sum_{k=0}^{\infty} (-1)^k \, a_{i,2i+1+k} \, \hat{u}^k, \tag{S2}$$

which converges for $|\hat{u}| < R$, where R is the radius of the largest disc about the origin on which Θ is analytic and $1 + 2\Theta$ has no zero, so that the singularity of b_i at

the mean is removable, with $b_i \to a_{i,2i+1}$ as $\bar{x} \to \mu$; (S1) then holds for all $\bar{x}$, with b_i given by its closed form away from the mean and by (S2) near it, and at $\bar{x} = \mu$ it reads $\frac{1}{2} - \phi(0) \sum_{i=0}^{M-1} a_{i,2i+1} n^{-i-1/2}$.

For $M = 1$, (S1) is the familiar $\Phi(\hat{w}_n) + \phi(\hat{w}_n)(\hat{w}_n^{-1} - \hat{u}_n^{-1})$. The coefficient b_i is Lugannani and Rice's; its two parts are individually singular at the mean, and (S2) shows that the singularities cancel to all orders. The proof is one line from (30), for $|\hat{u}| < R$: since $\hat{w}^2 = 2\sum_{k=2}^{\infty}(-1)^k \theta_k \hat{u}^k = \hat{u}^2\{1 + 2\Theta(-\hat{u})\}$,

$$\frac{h_i}{\hat{w}^{2i+1}} = \frac{h_i\{1 + 2\Theta(-\hat{u})\}^{-i-1/2}}{\hat{u}^{2i+1}} = \sum_{j=0}^{\infty} \frac{(-1)^j a_{i,j}}{\hat{u}^{2i+1-j}},$$

and the first sum in b_i is exactly the first $2i + 1$ terms of this series. The radius of convergence of Θ alone does not suffice: for $X = Z + N$ with Z standard normal and N Poisson with mean one, $K(s) = s^2/2 + e^s - 1$ and Θ is entire, but at $\hat{s} = 10$, where $\hat{u} = 10(1 + e^{10})^{1/2} \simeq 1484$, the function $1 + 2\Theta$ vanishes at a complex point of modulus about 1150, which bounds the radius of (S2) there; the series is used only near the mean, where $|\hat{u}|$ is small. In computation the series (S2) is used for $|\hat{u}|$ below a threshold and the closed form (S1) elsewhere; the closed form loses about $\log_{10}(1/\hat{u}^{2i+1})$ digits to cancellation at order i. For the first-order term, Yang and Kolassa (2002) expand $1/\hat{w} - 1/\hat{u}$ about the mean to first order in $\hat{w}$, with coefficients evaluated at the mean, for the same purpose and also for the double saddlepoint approximation; (S2) gives every term of every b_i, with coefficients evaluated at the saddlepoint. At the mean the value of the M-term approximation (S1) coincides with that of the Hauschildt–Robinson approximation with $p = 2M - 1$, equation (40) of the paper.

The derivative of (S1) is again a density expansion. Routledge and Tsao (1997) proved, under a uniform validity assumption, that the Lugannani–Rice expansion may be differentiated term by term to give Daniels' expansion, and hence that the derivative of its truncated version is a density approximation of the same order. In the present notation the result is an algebraic identity, proved below:

$$\sqrt{\hat{K}_2}\,\frac{\mathrm{d}b_i}{\mathrm{d}\bar{x}} = \hat{u}\, b_{i+1} - a_{i+1}, \tag{S3}$$

so that the derivative of the M-term approximation (S1) is

$$\frac{\mathrm{d}}{\mathrm{d}\bar{x}}\left\{\Phi(\hat{w}_n) - \phi(\hat{w}_n)\sum_{i=0}^{M-1}\frac{b_i}{n^{i+1/2}}\right\} = \frac{\sqrt{n}\,\phi(\sqrt{n}\hat{w})}{\sqrt{\hat{K}_2}}\left\{\sum_{i=0}^{M}\frac{a_i}{n^i} - \frac{\hat{u}\,b_M}{n^M}\right\}, \tag{S4}$$

Daniels' approximation with $M + 1$ terms minus a term of relative order n^{-M} that vanishes at the mean. For $M = 1$, with $\hat{u}b_1 = -\hat{u}^{-2} - 3\theta_3\hat{u}^{-1} + a_1 + \hat{u}\hat{w}^{-3}$, the right side of (S4) is

$$\frac{\sqrt{n}\,\phi(\sqrt{n}\hat{w})}{\sqrt{\hat{K}_2}}\left\{1 + \frac{1}{n}\left(\frac{1}{\hat{u}^2} + \frac{3\theta_3}{\hat{u}} - \frac{\hat{u}}{\hat{w}^3}\right)\right\}, \tag{S5}$$

the adjusted saddlepoint approximation of Routledge and Tsao (1997), which they found to be more accurate than Daniels' first-order approximation, in particular near the mean, where its correction tends to a_1/n and makes it of second order. The three terms of the correction are individually singular at the mean, a singularity inherited from (S1); near the mean $\hat{u}b_M$ has to be computed from the series (S2), with the same loss of digits in the closed form as for (S1).

PROOF OF THE DERIVATIVE IDENTITY (S3). Write $\mathcal{D} = \hat{K}_2^{1/2}\mathrm{d}/\mathrm{d}\bar{x}$. Since $\mathrm{d}\hat{s}/\mathrm{d}\bar{x} = 1/\hat{K}_2$, we have $\mathcal{D}\hat{u} = 1 + 3\theta_3\hat{u}$, $\hat{w}\mathcal{D}\hat{w} = \hat{u}$ and $\mathcal{D}\theta_k = (k+1)\theta_{k+1} - 3k\theta_3\theta_k$. The $\hat{w}$-terms of both sides of (S3) agree, because $\mathcal{D}(h_i\hat{w}^{-2i-1}) = h_{i+1}\hat{u}\hat{w}^{-2i-3}$. Comparing the coefficients of $\hat{u}^{j-2i-2}$ in the remaining terms, and using $a_{i+1,2i+2} = a_{i+1}$ and $a_{i,0} = h_i$, (S3) is equivalent to

$$a_{i+1,j} = 3(2i+2-j)\theta_3 a_{i,j-1} - \mathcal{D}a_{i,j-1} - (2i+1-j)a_{i,j}, \qquad j \geq 0, \tag{S6}$$

with $a_{i,-1} = 0$. Let $P(s) = 1 + 2\Theta(s)$ and $A_i(s) = \sum_{j=0}^{\infty} a_{i,j}s^j = h_i P^{-\nu}$, $\nu = i + 1/2$, by (30). From $\mathcal{D}\theta_k$,

$$s\mathcal{D}P = sP' + 2P - 2 - 3\theta_3 s^2 P' - 6\theta_3 sP. \tag{S7}$$

Multiplying (S6) by s^j and summing, its left side minus its right side has generating function

$$\begin{aligned} &A_{i+1} - 3\theta_3\{(2i+1)sA_i - s^2A_i'\} + s\mathcal{D}A_i + (2i+1)A_i - sA_i' \\ &\quad = \nu h_i P^{-\nu-1}(-2 + 2P + sP' - s\mathcal{D}P - 6\theta_3 sP - 3\theta_3 s^2 P') = 0, \end{aligned} \tag{S8}$$

by $A_{i+1} = -2\nu h_i P^{-\nu-1}$, $A_i' = -\nu h_i P^{-\nu-1}P'$, $\mathcal{D}A_i = -\nu h_i P^{-\nu-1}\mathcal{D}P$ and (S7). Finally, differentiating (S1) with $\mathcal{D}\hat{w} = \hat{u}/\hat{w}$ and $\hat{u}(1/\hat{w} + b_0) = 1$ gives (S4). □

PROOF OF THEOREM S1. The series (S2) and the behavior at the mean were proved above; it remains to prove the expansion (S1) with its error bound and the closed form of b_i. By the reflection $\bar{X} \to -\bar{X}$ it suffices to take $\hat{s} > 0$, and we write $\lambda(s) = K(s) - s\bar{x}$, so that $\lambda(\hat{s}) = -\hat{w}^2/2$, $\lambda'(\hat{s}) = 0$ and $\lambda''(\hat{s}) = \hat{K}_2$.

The change of variable. Define $w = w(s)$ by

$$\tfrac{1}{2}(w - \hat{w})^2 = \lambda(s) - \lambda(\hat{s}), \qquad \operatorname{sgn}(w - \hat{w}) = \operatorname{sgn}(s - \hat{s}) \quad \text{for real } s. \tag{S9}$$

Since λ is strictly convex on the real domain of K with its minimum at $\hat{s}$, $w(s)$ is a real analytic increasing bijection of the real domain onto an interval, analytic on a complex neighborhood of it with $w'(\hat{s}) = \hat{K}_2^{1/2}$, and the neighborhood and the bounds on w' and $1/w'$ can be chosen uniformly for $\hat{s}$ in a compact subset T of the domain (Jensen, 1995, Section 3.3). At $s = 0$, $\lambda(0) = 0$ gives $(w - \hat{w})^2 = \hat{w}^2$, and the sign condition selects $w(0) = 0$. Hence the function

$$\psi(w) = \frac{1}{s}\frac{\mathrm{d}s}{\mathrm{d}w} - \frac{1}{w} \tag{S10}$$

is analytic on a neighborhood of the real w-interval, including $w = 0$, where the simple poles of the two terms cancel because $s = 0$ is a simple zero of $w(s)$; it is the function expanded by Lugannani and Rice (1980).

The expansion. For $\hat{s} > 0$ the inversion integral (33) may be taken along $\operatorname{Re} s = \hat{s}$. Let $\delta > 0$ be such that the map (S9) is analytic and invertible, with ψ bounded, on $\{s : |s - \hat{s}| \leq 2\delta\}$ for all $\hat{s} \in T$, and split the contour at $|\operatorname{Im} s| = \delta$. On the outer part, by the regularity condition (1) as in the proof of Theorem 1 of the paper, $|e^{n\lambda(s)}| \leq e^{n\lambda(\hat{s})}\rho^{n-n_0}|M_{\hat{s}}(y)|^{n_0}$ with $\rho < 1$, so that this part is exponentially small relative to $e^{-n\hat{w}^2/2}$, uniformly on T. On the inner part substitute w: with $\lambda(s) = w^2/2 - \hat{w}w$ by (S9),

$$\frac{1}{2\pi\mathrm{i}}\int_{\hat{s}-\mathrm{i}\delta}^{\hat{s}+\mathrm{i}\delta} e^{n\lambda(s)}\frac{\mathrm{d}s}{s} = \frac{1}{2\pi\mathrm{i}}\int_{w(\hat{s}-\mathrm{i}\delta)}^{w(\hat{s}+\mathrm{i}\delta)} e^{n(w^2/2-\hat{w}w)}\Big\{\frac{1}{w} + \psi(w)\Big\}\mathrm{d}w, \tag{S11}$$

over the image of the inner contour, which is a curve from $w(\hat{s} - \mathrm{i}\delta)$ to $w(\hat{s} + \mathrm{i}\delta)$ through $\hat{w}$ on which $\operatorname{Re} w$ stays within a fixed distance of $\hat{w}$ and $|\operatorname{Im} w|$ is comparable to $|\operatorname{Im} s|$. Deform it to the segment $\operatorname{Re} w = \hat{w}$, $|\operatorname{Im} w| \leq \delta'$, which does not cross the pole at 0 since $\hat{w} > 0$; the

two connecting pieces contribute $O(e^{-n(\hat{w}^2+c)/2})$ for some $c>0$, since $\mathrm{Re}(w^2/2-\hat{w}w) = -\hat{w}^2/2+\{(\mathrm{Re}\,w-\hat{w})^2-(\mathrm{Im}\,w)^2\}/2$ and $|\mathrm{Im}\,w|$ is bounded below on them. The term $1/w$ in (S11), completed to the whole line at an exponentially small cost and evaluated by (34) with $v=\sqrt{n}w$, gives $I_0(\hat{w}_n)=\Phi(-\hat{w}_n)$. For the term with ψ, write $w=\hat{w}+\mathrm{i}y/\sqrt{n}$:

$$\frac{1}{2\pi\mathrm{i}}\int_{\hat{w}-\mathrm{i}\delta'}^{\hat{w}+\mathrm{i}\delta'} e^{n(w^2/2-\hat{w}w)}\psi(w)\,\mathrm{d}w = \frac{e^{-n\hat{w}^2/2}}{2\pi\sqrt{n}}\int_{-\delta'\sqrt{n}}^{\delta'\sqrt{n}} e^{-y^2/2}\psi(\hat{w}+\mathrm{i}y/\sqrt{n})\,\mathrm{d}y. \tag{S12}$$

Expanding ψ in its Taylor series at $\hat{w}$, which converges on $|w-\hat{w}|\le\delta'$ with coefficients bounded uniformly on T by Cauchy's estimates, and integrating term by term with $\int_{-\infty}^{\infty} y^l e^{-y^2/2}\mathrm{d}y = (2\pi)^{1/2}E(Y^l)$, Y standard normal, the term of degree l contributes

$$\phi(\hat{w}_n)\,\psi^{(l)}(\hat{w})\,\mathrm{i}^l E(Y^l)\,\frac{n^{-(l+1)/2}}{l!},$$

which vanishes for odd l and equals $\phi(\hat{w}_n)\psi^{(2i)}(\hat{w})h_i n^{-i-1/2}/(2i)!$ for $l=2i$, since $\mathrm{i}^{2i}E(Y^{2i})=(-1)^i(2i-1)!!=h_i$; the remainder after the terms of degree below $2M$ is bounded by a constant times $n^{-M}\int_{-\infty}^{\infty}|y|^{2M}e^{-y^2/2}\mathrm{d}y$, and extending the retained terms to the whole line costs exponentially little. Altogether

$$\mathbb{P}(\bar{X}>\bar{x}) = \Phi(-\hat{w}_n)+\phi(\hat{w}_n)\sum_{i=0}^{M-1}\frac{\tilde{b}_i}{n^{i+1/2}}+\phi(\hat{w}_n)O(n^{-M-1/2}), \tag{S13}$$

with

$$\tilde{b}_i = \frac{h_i\psi^{(2i)}(\hat{w})}{(2i)!},$$

uniformly on T. Since $\Phi(-\hat{w}_n)$ is of exact order $\phi(\hat{w}_n)/(1+\hat{w}_n)$, the error is $O(n^{-M})$ relative to the leading term, uniformly on T, and taking complements gives (S1) with $b_i=\tilde{b}_i$. Explicitly, by (S10) and $(\mathrm{d}/\mathrm{d}w)^{2i}w^{-1}=(2i)!\,w^{-2i-1}$,

$$b_i = \frac{h_i}{(2i)!}\Big(\frac{\mathrm{d}}{\mathrm{d}w}\Big)^{2i}\Big\{\frac{1}{s}\frac{\mathrm{d}s}{\mathrm{d}w}\Big\}\Big|_{w=\hat{w}} - \frac{h_i}{\hat{w}^{2i+1}}. \tag{S14}$$

This is Lugannani and Rice's form of the coefficients, and the removable singularity at the mean is visible in it: ψ is analytic at $w=0$.

The closed form. It remains to show that, for $\hat{s}>0$ fixed, the first term of (S14) equals the first sum in (S1), $\sum_{j=0}^{2i}(-1)^j a_{i,j}\hat{u}^{-(2i+1-j)}$. Both (S13) and the Hauschildt–Robinson expansion (36) of the paper, the latter for every p, are asymptotic expansions of $\mathbb{P}(\bar{X}>\bar{x})$ as $n\to\infty$. Write $I_m(u)=\phi(u)\tilde{I}_m(u)$ with $\tilde{I}_m(u)=\int_0^\infty e^{-ut}\mathrm{He}_m(t)e^{-t^2/2}\mathrm{d}t$, by (34), and note that $E_n\phi(\hat{u}_n)=\phi(\hat{w}_n)$ by (32). By Watson's lemma, as $u\to\infty$,

$$\begin{aligned}\tilde{I}_m(u) &\sim \sum_{l=0}^{\infty}\frac{g_{m,l}}{u^{l+1}},\\ g_{m,l} &= \Big(\frac{\mathrm{d}}{\mathrm{d}t}\Big)^l\big\{\mathrm{He}_m(t)e^{-t^2/2}\big\}\Big|_{t=0} = (-1)^m\Big(\frac{\mathrm{d}}{\mathrm{d}t}\Big)^{l+m}e^{-t^2/2}\Big|_{t=0},\end{aligned} \tag{S15}$$

since $\mathrm{He}_m(t)e^{-t^2/2}=(-1)^m(\mathrm{d}/\mathrm{d}t)^m e^{-t^2/2}$; the derivative of $e^{-t^2/2}$ at 0 of order $2k$ is h_k and of odd order is 0. Substituting $u=\hat{u}_n$ into (36), the term with (r,m) and l carries $n^{-m/2}n^{-(l+1)/2}$, so the coefficient of $\phi(\hat{w}_n)n^{-i-1/2}$ collects the terms with $l=2i-m$, for which $l+m+2r=2i+2r$ is even and $g_{m+2r,2i-m}=(-1)^m h_{i+r}$:

$$\sum_{m=0}^{2i}\sum_{r=0}^{m} d_{r,m}\frac{(-1)^m h_{i+r}}{\hat{u}^{2i+1-m}} = \sum_{m=0}^{2i}\frac{(-1)^m a_{i,m}}{\hat{u}^{2i+1-m}}, \tag{S16}$$

by (29), the term $m = r = 0$ being that of $I_0(\hat{u}_n) = \Phi(-\hat{u}_n)$ and $d_{0,m} = 0$ for $m > 0$. The terms of (36) with $m > p$ are absorbed in its error, which is $O(n^{-(p+1)/2})$ relative to the leading term $E_n I_0(\hat{u}_n)$, itself of exact order $\phi(\hat{w}_n)n^{-1/2}$; choosing $p \geq 2i + 1$ shows that (S16) is the coefficient of $\phi(\hat{w}_n)n^{-i-1/2}$ in an asymptotic expansion of $\mathbb{P}(\bar{X} > \bar{x})$ in the sequence $\{\phi(\hat{w}_n)n^{-i-1/2}\}$. On the other side, $\Phi(-\hat{w}_n) \sim \phi(\hat{w}_n)\sum_{i=0}^{\infty} h_i \hat{w}_n^{-2i-1}$ by the expansion of Mills' ratio, so (S13) is an asymptotic expansion in the same sequence with coefficients $\tilde{b}_i + h_i \hat{w}^{-2i-1}$. Asymptotic expansions in a given sequence are unique, so the first term of (S14) equals (S16), which is the closed form of (S1). The closed form holds for $\hat{s} < 0$ by the reflection, and the series (S2) extends it across the mean.

The uniformity is claimed only away from the mean, where the contour deformation above is possible: the pulled-back segment $\operatorname{Re} w = \hat{w}$ deviates from $\operatorname{Re} s = \hat{s}$ by $O(|\operatorname{Im} s|^2)$ and must stay to the right of the pole $s = 0$, which bounds δ in terms of $\hat{s}$. At the mean itself the M-term approximation (S1) coincides with the Hauschildt–Robinson approximation with $p = 2M - 1$, equation (40) of the paper, and carries the bound of Theorem 1, which holds up to and including $\hat{s} = 0$; a bound that is uniform in a neighborhood of the mean is known for $M = 1$ (Jensen, 1995, formula (3.3.9)) and is not claimed here for $M > 1$. □

S2. Noncentral chi-squared and *F* distributions. The setting is that of the second group of examples of Section 5 of the paper: the "n" of the expansions is the degrees of freedom or the noncentrality rather than a sample size, the expansions are written with $n = 1$ and the cumulant generating function of the variable itself, the relative error of a density approximation $\hat{f}$ is $(\hat{f} - f)/f$ and that of an approximation $\hat{F}$ to a distribution function is $(\hat{F} - F)/\min(F, 1 - F)$, and the tables summarize the absolute values of these errors over grids. All computations were done in MATLAB in double precision, with the integrals I_m evaluated by quadrature when $|\hat{u}_n| > 2.5$ (Appendix C of the paper). For the noncentral chi-squared distribution the setting corresponds to the scaling family $\chi^2_{L\nu_0}(L\lambda_0)$, the sum of L independent $\chi^2_{\nu_0}(\lambda_0)$ variables, so that the degrees of freedom and the noncentrality play the role of n when they grow together, and the noncentral F distributions inherit this structure from their two components.

S2.1. *Noncentral chi-squared distribution.* Let $X \sim \chi^2_\nu(\lambda)$, with $K(s) = -\frac{\nu}{2}\log(1 - 2s) + \lambda s/(1 - 2s)$, $s < 1/2$. The saddlepoint solves a quadratic equation, and the derivatives of K are available in closed form,

$$K^{(k)}(s) = \frac{2^{k-1}(k-1)!\,\nu}{(1-2s)^k} + \frac{2^{k-1}k!\,\lambda}{(1-2s)^{k+1}}. \tag{S17}$$

We evaluated the approximations of orders one to six of the density (Corollary 1 of the paper) and of the distribution function (Corollary 2, $p = 2M - 1$) for $\nu = 1, \ldots, 200$, $\lambda = 1, 2, 5, 10, 20, 50, 100, 500, 1000, 5000$, and at the nominal quantiles of probability $0.001, 0.01, 0.02, \ldots, 0.99, 0.999$ of each distribution, the values of MATLAB's `ncx2inv` rounded to four significant digits, at which the exact values and the errors are evaluated: 202,000 points. Exact values were computed in Mathematica in high precision. Table S1 reports the median and the maximum of the relative error, the latter both over all ν and over $\nu \geq 10$; for the distribution function the error is relative to $\min(F, 1 - F)$.

Three features of Table S1 recur, to varying degrees, in the other examples, here and in the paper. First, the median error falls by roughly two orders of magnitude per order of the expansion, from 10^{-3}–10^{-5} at order one to 10^{-13}–10^{-16} at order six, close to the rounding level of double precision; for $\lambda \geq 100$ the maximum error behaves in the same way. Second, the expansion is asymptotic, not convergent: for $\lambda \leq 10$ and the smallest ν the maximum error stops improving after order three or four and eventually grows, while the maximum

TABLE S1

Noncentral chi-squared: relative errors of the approximations of order M over $\nu = 1, \ldots, 200$ and 101 quantiles. For each λ: median, maximum over all ν, maximum over $\nu \geq 10$ (notation $a_k = a \times 10^k$).

	$\lambda = 1$			$\lambda = 10$			$\lambda = 100$			$\lambda = 1000$		
M	med	max	$\max_{\nu \geq 10}$	med	max	$\max_{\nu \geq 10}$	med	max	$\max_{\nu \geq 10}$	med	max	$\max_{\nu \geq 10}$
Distribution function												
1	6.9_{-4}	6.4_{-2}	1.2_{-2}	6.5_{-4}	7.9_{-2}	7.9_{-3}	3.0_{-4}	1.1_{-3}	1.0_{-3}	4.0_{-5}	1.2_{-4}	1.2_{-4}
2	1.5_{-6}	2.2_{-2}	7.9_{-4}	1.8_{-6}	1.4_{-2}	7.6_{-4}	8.7_{-7}	6.1_{-6}	5.9_{-6}	1.9_{-8}	5.0_{-8}	4.7_{-8}
3	1.2_{-8}	2.0_{-2}	7.1_{-5}	1.2_{-8}	2.6_{-2}	2.4_{-5}	2.5_{-9}	1.1_{-7}	1.1_{-7}	1.3_{-11}	5.8_{-11}	5.8_{-11}
4	9.3_{-11}	2.5_{-2}	8.3_{-6}	1.6_{-10}	1.2_{-2}	8.3_{-6}	9.8_{-12}	3.6_{-9}	3.5_{-9}	1.3_{-14}	1.2_{-13}	1.2_{-13}
5	1.6_{-12}	4.9_{-2}	2.2_{-6}	2.4_{-12}	6.0_{-2}	1.6_{-6}	1.3_{-13}	1.4_{-10}	1.3_{-10}	1.7_{-15}	6.3_{-14}	6.3_{-14}
6	1.9_{-14}	1.1_{-1}	4.2_{-7}	5.7_{-14}	2.8_{-3}	1.9_{-7}	2.6_{-15}	6.3_{-12}	4.8_{-12}	1.7_{-15}	6.3_{-14}	6.3_{-14}
Density												
1	1.7_{-3}	1.9_{-1}	1.8_{-2}	1.7_{-3}	1.8_{-1}	1.8_{-2}	1.5_{-3}	5.4_{-3}	4.6_{-3}	3.3_{-4}	4.2_{-4}	4.1_{-4}
2	1.4_{-6}	3.4_{-2}	2.0_{-4}	1.6_{-6}	3.3_{-2}	1.9_{-4}	1.8_{-6}	2.5_{-5}	1.9_{-5}	9.8_{-8}	1.4_{-7}	1.4_{-7}
3	2.1_{-8}	3.5_{-2}	4.4_{-5}	2.8_{-8}	2.6_{-2}	4.4_{-5}	3.3_{-9}	3.1_{-7}	2.2_{-7}	8.0_{-11}	1.4_{-10}	1.4_{-10}
4	3.6_{-11}	2.9_{-2}	2.6_{-6}	1.8_{-10}	2.3_{-2}	5.1_{-6}	5.6_{-11}	6.5_{-9}	3.9_{-9}	1.0_{-13}	2.3_{-13}	2.1_{-13}
5	2.4_{-12}	6.8_{-2}	1.3_{-6}	5.2_{-12}	6.1_{-2}	8.6_{-7}	1.0_{-12}	1.8_{-10}	9.6_{-11}	6.2_{-16}	1.2_{-14}	1.2_{-14}
6	2.4_{-15}	1.2_{-1}	3.1_{-7}	1.5_{-13}	3.6_{-2}	3.5_{-7}	1.9_{-14}	6.4_{-12}	3.0_{-12}	6.0_{-16}	1.3_{-14}	1.3_{-14}

over $\nu \geq 10$ continues to decrease with the order, so the large errors at high order arise from fewer than ten degrees of freedom. Third, the density and distribution function expansions behave alike, their formal derivatives agreeing to the order of the expansion (Section S1).

S2.2. *Singly noncentral F distribution.* Let $F = (Q_1/m)/(Q_2/n)$ with $Q_1 \sim \chi^2_m(\lambda)$ and $Q_2 \sim \chi^2_n$ independent. Then $\mathbb{P}(F \leq f) = \mathbb{P}(W \leq 0)$ with $W = Q_1 - cQ_2$, $c = mf/n$, and $K_W(s) = K_{Q_1}(s) + K_{Q_2}(-cs)$ on $-1/(2c) < s < 1/2$; the saddlepoint again solves a quadratic. The distribution function is approximated by Corollary 2 with $\bar{x} = 0$, and the density by Corollary 5 with $g(s) = K'_{Q_2}(-cs)$ and $f_F(f) = (m/n) f_R(c)$, $R = Q_1/Q_2$. Butler and Paolella (2002) use the first- and second-order Lugannani–Rice approximations for this distribution and report relative errors of about 0.03% for the latter; our orders one and two correspond to theirs, up to the choice of the Hauschildt–Robinson form. We evaluated orders one to six for $m, n = 1, \ldots, 100$, the ten values of λ above and the same 101 nominal quantiles, from `ncfinv` rounded to four significant digits: 1.01×10^7 points. Exact values were computed in Mathematica in high precision. Table S2 is organized like Table S1, with the maximum over $\min(m, n) \geq 10$ third.

The pattern is as before, but the maximum errors are now dominated by the small denominator degrees of freedom: the large parameter is $\min(m, n)$, and with $n \leq 2$ the largest error of the distribution function on this grid is smallest at order three, between 0.04 and 0.09, and does not decrease at higher orders, where it stays between 0.06 and 0.3. Over $\min(m, n) \geq 10$ the maximum error improves steadily with the order, to 10^{-6}–10^{-7} at order six, and the median over all (m, n) reaches 8×10^{-11}–4×10^{-13}. The noncentrality parameter matters less than for the chi-squared distribution, because it enters only the numerator.

S2.3. *Doubly noncentral F distribution.* Let $F = (Q_1/m)/(Q_2/n)$ with $Q_1 \sim \chi^2_m(\lambda_1)$ and $Q_2 \sim \chi^2_n(\lambda_2)$ independent. The distribution has no elementary closed form; its distribution function is a double Poisson mixture of incomplete beta functions, and the second-order saddlepoint approximations of Butler and Paolella (2002), which they found to run thousands of times faster than numerical inversion, are the usual way to compute it (Butler, 2007, Section 12.1). The expansions are exactly those of Section S2.2 with $K_W(s) = K_{Q_1}(s) + K_{Q_2}(-cs)$ for both noncentral components, and the saddlepoint is again explicit: $K'_W(s) = 0$ reduces to a cubic equation, whose middle root is the saddlepoint (Butler and

TABLE S2

Singly noncentral F: relative errors of the approximations of order M over $m, n = 1, \ldots, 100$ and 101 quantiles. For each λ: median, maximum over all (m, n), maximum over $\min(m, n) \geq 10$ (notation $a_k = a \times 10^k$).

	$\lambda = 1$			$\lambda = 10$			$\lambda = 100$			$\lambda = 1000$		
M	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$
Distribution function												
1	4.4_{-3}	2.0_{-1}	5.0_{-2}	3.9_{-3}	2.6_{-1}	4.8_{-2}	2.6_{-3}	1.2_{-1}	1.9_{-2}	1.8_{-3}	1.2_{-1}	1.3_{-2}
2	1.3_{-5}	9.4_{-2}	1.9_{-3}	1.4_{-5}	1.3_{-1}	2.3_{-3}	6.8_{-6}	9.2_{-2}	1.2_{-3}	6.0_{-6}	9.1_{-2}	8.0_{-4}
3	7.6_{-7}	5.3_{-2}	1.5_{-4}	6.1_{-7}	8.6_{-2}	8.4_{-5}	1.3_{-7}	4.0_{-2}	7.5_{-5}	1.1_{-7}	3.8_{-2}	4.8_{-5}
4	1.2_{-8}	6.4_{-2}	2.8_{-5}	1.8_{-8}	7.2_{-2}	3.6_{-5}	1.4_{-9}	6.2_{-2}	9.1_{-6}	1.7_{-9}	6.2_{-2}	7.2_{-6}
5	1.0_{-9}	1.2_{-1}	3.4_{-6}	9.6_{-10}	1.2_{-1}	1.8_{-6}	7.7_{-11}	6.6_{-2}	1.4_{-6}	5.4_{-11}	6.4_{-2}	9.0_{-7}
6	2.6_{-11}	2.0_{-1}	1.2_{-6}	7.7_{-11}	3.0_{-1}	1.4_{-6}	1.5_{-12}	1.4_{-1}	2.7_{-7}	1.4_{-12}	1.4_{-1}	2.1_{-7}
Density												
1	6.6_{-3}	2.8_{-1}	2.6_{-2}	6.9_{-3}	2.7_{-1}	2.7_{-2}	5.0_{-3}	1.7_{-1}	1.8_{-2}	3.4_{-3}	1.7_{-1}	1.7_{-2}
2	2.3_{-5}	6.5_{-2}	3.9_{-4}	1.6_{-5}	5.7_{-2}	3.0_{-4}	8.5_{-6}	2.9_{-2}	1.7_{-4}	5.6_{-6}	2.8_{-2}	1.6_{-4}
3	8.9_{-7}	4.3_{-2}	4.7_{-5}	1.2_{-6}	4.3_{-2}	6.1_{-5}	1.8_{-7}	1.2_{-2}	2.1_{-5}	1.7_{-7}	1.2_{-2}	2.1_{-5}
4	8.0_{-9}	4.4_{-2}	3.4_{-6}	1.3_{-8}	2.3_{-2}	5.1_{-6}	2.7_{-9}	1.3_{-2}	6.9_{-7}	6.6_{-10}	1.3_{-2}	6.2_{-7}
5	9.2_{-10}	7.5_{-2}	1.3_{-6}	2.0_{-9}	6.8_{-2}	1.3_{-6}	1.3_{-10}	8.4_{-3}	2.5_{-7}	7.6_{-11}	8.4_{-3}	2.4_{-7}
6	9.0_{-12}	1.5_{-1}	2.3_{-7}	7.2_{-11}	7.4_{-2}	3.4_{-7}	2.0_{-12}	2.1_{-2}	1.2_{-8}	3.7_{-13}	2.1_{-2}	1.2_{-8}

Paolella, 2002), which we refine by Newton's method. We evaluated orders one to six for all $m, n = 1, \ldots, 100$, all sixteen pairs $(\lambda_1, \lambda_2) \in \{1, 10, 100, 1000\}^2$, and the quantiles of probability 0.001, $0.01, 0.02, \ldots, 0.99$ and 0.999, located by Newton's method on the exact distribution function to a relative tolerance of 10^{-6} in the smaller tail: 1.6×10^7 points. The exact values of F, $1 - F$ and the density were computed in double precision from the double Poisson mixture, with the incomplete beta functions generated by recurrences in their stable direction and the Poisson series truncated at a negligible remainder. They agree with values computed in Mathematica with 40-digit working precision at 2816 points of a smaller grid to within 1.2×10^{-12}, and with one-dimensional numerical integration over Q_2 at 800 random points of the grid to within 6×10^{-13}, the larger differences occurring when λ_1 or λ_2 is 1000; errors below about 10^{-12} are therefore at the accuracy of the exact values. Since $1/F$ is doubly noncentral F with parameters $(n, m, \lambda_2, \lambda_1)$ and $\{1/F \leq 1/f\} = \{-W/f \leq 0\}$, a scaling of W by a negative constant under which the expansions are invariant, the pairs (λ_1, λ_2) and (λ_2, λ_1) give the same summaries, as the computations confirm. Table S3 shows the four pairs with $\lambda_1 = \lambda_2$ and is organized like Table S2.

The pattern is that of Table S2. Over $\min(m, n) \geq 10$ the maximum error falls steadily with the order, to about 10^{-6} for $\lambda_1 = \lambda_2 \leq 10$ and to 3×10^{-11} for $(100, 100)$, where the noncentralities act as the large parameter; for $(1000, 1000)$ the approximation of order four already reaches the accuracy of the exact values. The medians decrease with the order up to order six for every pair except $(1000, 1000)$, where they reach that accuracy at order four. For $\lambda_1 = \lambda_2 \leq 10$ the maxima over all (m, n) are attained at $m = n = 1$, where the expansion stops improving after order two to four, as in Section S2.2; excluding $\min(m, n) = 1$, the maximum error of order six for $(1, 1)$ is 7×10^{-3}, and over $\min(m, n) \geq 5$ it is 4×10^{-5}. The twelve pairs with $\lambda_1 \neq \lambda_2$, six of them distinct, behave in the same way: at order six their medians are between 2×10^{-13} and 7×10^{-11} and their maxima over $\min(m, n) \geq 10$ between 4×10^{-12} and 2×10^{-6}. The rows $M = 1, 2$ are the approximations of Butler and Paolella (2002) in Hauschildt–Robinson form.

S3. The approximations of Wang (1990) and of Kolassa and Li (2010): details.

S3.1. *Wang's example with n doubling.* In Wang's example (Section 5.1 of the paper), Table S4 fixes three points and doubles n from 10 to 160, so that the per-observation saddle-

TABLE S3

Doubly noncentral F: relative errors of the approximations of order M over $m, n = 1, \ldots, 100$ and 101 quantiles. For each (λ_1, λ_2): median, maximum over all (m, n), maximum over $\min(m, n) \geq 10$. Entries below about 10^{-12} are at the accuracy of the exact values (notation $a_k = a \times 10^k$).

	$(\lambda_1, \lambda_2) = (1, 1)$			$(10, 10)$			$(100, 100)$			$(1000, 1000)$		
M	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$	med	max	$\max_{\geq 10}$
Distribution function												
1	4.4_{-3}	1.1_{-1}	4.8_{-2}	3.5_{-3}	1.0_{-1}	3.3_{-2}	1.2_{-3}	5.8_{-3}	5.4_{-3}	1.6_{-4}	5.6_{-4}	5.6_{-4}
2	1.3_{-5}	4.2_{-2}	1.9_{-3}	1.6_{-5}	1.9_{-2}	1.5_{-3}	3.7_{-6}	8.1_{-5}	6.8_{-5}	8.0_{-8}	7.1_{-7}	7.0_{-7}
3	7.8_{-7}	2.9_{-2}	1.4_{-4}	5.0_{-7}	3.6_{-2}	2.4_{-5}	2.0_{-8}	1.8_{-6}	1.2_{-6}	8.8_{-11}	1.3_{-9}	1.3_{-9}
4	1.3_{-8}	6.9_{-2}	2.8_{-5}	2.4_{-8}	1.2_{-2}	2.0_{-5}	2.2_{-10}	5.2_{-8}	3.0_{-8}	2.7_{-13}	3.2_{-12}	3.0_{-12}
5	1.1_{-9}	1.0_{-1}	3.4_{-6}	8.7_{-10}	7.4_{-2}	1.8_{-6}	6.6_{-12}	2.0_{-9}	9.1_{-10}	3.8_{-13}	7.6_{-13}	7.6_{-13}
6	2.1_{-11}	2.3_{-1}	1.2_{-6}	1.5_{-10}	4.4_{-2}	8.6_{-7}	1.9_{-13}	1.0_{-10}	3.2_{-11}	3.8_{-13}	7.6_{-13}	7.6_{-13}
Density												
1	6.6_{-3}	3.0_{-1}	2.6_{-2}	7.1_{-3}	1.9_{-1}	2.6_{-2}	4.2_{-3}	8.3_{-3}	7.0_{-3}	7.0_{-4}	7.6_{-4}	7.5_{-4}
2	2.4_{-5}	4.0_{-2}	4.2_{-4}	1.3_{-5}	3.3_{-2}	2.3_{-4}	1.2_{-5}	5.7_{-5}	4.0_{-5}	4.0_{-7}	4.8_{-7}	4.6_{-7}
3	8.7_{-7}	5.8_{-2}	4.7_{-5}	1.5_{-6}	2.8_{-2}	4.8_{-5}	8.4_{-8}	1.1_{-6}	6.1_{-7}	6.3_{-10}	8.4_{-10}	8.0_{-10}
4	8.8_{-9}	3.7_{-2}	4.1_{-6}	3.5_{-8}	2.3_{-2}	6.2_{-6}	7.0_{-10}	3.4_{-8}	1.5_{-8}	2.0_{-12}	2.9_{-12}	2.7_{-12}
5	7.8_{-10}	7.1_{-2}	1.3_{-6}	2.7_{-9}	6.7_{-2}	1.0_{-6}	2.3_{-11}	1.4_{-9}	4.8_{-10}	3.9_{-13}	8.0_{-13}	8.0_{-13}
6	8.5_{-12}	1.3_{-1}	2.3_{-7}	3.7_{-10}	3.4_{-2}	3.4_{-7}	7.7_{-13}	7.5_{-11}	1.9_{-11}	3.9_{-13}	8.0_{-13}	8.0_{-13}

point is fixed and the rates of Theorem 1 of the paper can be read off. The ratio of successive errors of order one approaches the $1/2$ of an $O(n^{-1})$ error, and those of orders two to five fall by factors that approach 2^{-M}, until at order five they reach the accuracy of the exact values, 10^{-13}. At the mean, where both components of the saddlepoint vanish and Wang's approximation is singular, the errors fall at the same rates, as Theorem 1 asserts on the closed orthant. The relative error of Wang's approximation decreases slowly at $(-0.5, -0.5)$, from 1.2% to 0.22%, roughly like $n^{-1/2}$, and not at all at $(-1.0, 2.0)$, where it stays at 5%. This is consistent with its derivation: the error term of exponential order times $O(n^{-1})$ that Kolassa (2003, Section 4) states for it controls the relative error in the moderate-deviation region $\hat{s} = O(n^{-1/2})$, and at a fixed point in two dimensions it is of the same order as the leading term, as discussed after Theorem 1.

S3.2. *Kolassa and Li's example with n doubling.* In Kolassa and Li's example (Section 5.1 of the paper), Table S5 fixes y at $(3.0, 3.0)$, $(3.5, 4.0)$ and $(2.5, 3.0)$ and doubles n from 5 to 160. The expansion behaves as in Table S4: the error of order one settles to halving with each doubling, those of orders two to five fall by factors that approach 2^{-M}, and the same holds at $(2.5, 3.0)$, where $\hat{s}_1 = 0$ and the approximation of Kolassa and Li is singular; there the order-one error changes sign between $n = 80$ and $n = 160$, and settles on its rate only beyond. The relative error of their approximation does not decrease: at $(3.5, 4.0)$ it stays between 0.87% and 0.98% (in the reversed order of the coordinates it passes from -0.22% to 0.80%), and at $(3.0, 3.0)$, where the model is symmetric in its two coordinates and the two orders coincide, it passes from -0.76% through zero to 0.19%. Their approximation is thus the more accurate at $n = 5$, where our order one is a first-order approximation of Hauschildt–Robinson type with five observations, but the order-one expansion overtakes it as n grows, at $n = 20$ at the second point and at $n = 160$ at the first, and order two is more accurate at every n of the table. Over this range the relative error of their approximation shows no sign of the $O(n^{-1})$ decay that Kolassa and Li (2010) state for it, uniformly on compact sets of $\bar{y}$ (their Section 1); their derivation absorbs terms whose relative contribution is $O(n^{-1/2})$ (the remark after their equation (2.11)), so the range $n \leq 160$ may be pre-asymptotic for it. Our implementation reproduces the eight values they print to the digits given, with the second derivative of the nested signed root by central differences (Section S3.4); this validates the

TABLE S4

Wang's example with the point fixed and n doubling: $F(x,y)=\mathbb{P}(\bar{X}\le x,\bar{Y}\le y)$ and relative errors $(\hat{F}-F)/F$ of Wang's approximation (W, our implementation of his Theorem 1, in the order of the coordinates that his rule selects; — at the mean, where it is singular) and of the expansions of orders 1 to 5 (notation $a_k=a\times 10^k$). At $(-0.5,-0.5)$ the saddlepoint of $(-\bar{X},-\bar{Y})$ is in the open orthant, at $(0,0)$ both of its components vanish, and at $(-1.0,2.0)$ one is negative and (17) of the paper is used. Entries below 10^{-12} in absolute value are at the accuracy of the exact values.

n	exact	W	1	2	3	4	5
$(x,y)=(-0.5,-0.5)$							
10	4.258299_{-2}	-1.2_{-2}	-1.4_{-2}	-7.2_{-5}	7.4_{-6}	2.0_{-7}	-3.3_{-8}
20	1.020789_{-2}	-8.1_{-3}	-8.2_{-3}	-4.1_{-5}	1.1_{-6}	2.5_{-8}	-1.1_{-9}
40	8.211831_{-4}	-5.5_{-3}	-4.3_{-3}	-1.8_{-5}	1.5_{-7}	2.9_{-9}	-3.0_{-11}
80	7.979907_{-6}	-3.5_{-3}	-2.1_{-3}	-6.8_{-6}	1.2_{-8}	3.0_{-10}	-5.2_{-14}
160	1.218616_{-9}	-2.2_{-3}	-9.3_{-4}	-2.2_{-6}	-3.2_{-10}	2.4_{-11}	9.6_{-14}
$(x,y)=(0.0,0.0)$							
10	3.644369_{-1}	—	-3.8_{-3}	3.9_{-6}	2.0_{-6}	4.6_{-8}	-1.0_{-8}
20	3.550428_{-1}	—	-1.9_{-3}	5.5_{-7}	2.5_{-7}	3.3_{-9}	-3.1_{-10}
40	3.485424_{-1}	—	-9.8_{-4}	7.4_{-8}	3.1_{-8}	2.2_{-10}	-9.8_{-12}
80	3.440169_{-1}	—	-5.0_{-4}	8.6_{-9}	3.9_{-9}	1.4_{-11}	-3.5_{-13}
160	3.408524_{-1}	—	-2.5_{-4}	5.7_{-10}	5.0_{-10}	9.0_{-13}	-2.7_{-14}
$(x,y)=(-1.0,2.0)$							
10	3.454341_{-3}	5.5_{-2}	-2.0_{-3}	-4.9_{-5}	-1.7_{-7}	2.7_{-8}	4.5_{-10}
20	5.320203_{-5}	5.3_{-2}	-3.0_{-4}	-1.1_{-5}	-8.6_{-8}	7.7_{-10}	2.9_{-11}
40	1.695472_{-8}	5.2_{-2}	2.3_{-4}	-1.9_{-6}	-1.1_{-8}	-6.5_{-12}	4.8_{-13}
80	2.361592_{-15}	5.1_{-2}	2.8_{-4}	-2.6_{-7}	-1.0_{-9}	-5.3_{-13}	2.2_{-16}
160	6.369616_{-29}	5.1_{-2}	1.9_{-4}	-3.9_{-8}	-9.4_{-11}	1.1_{-14}	-3.3_{-15}

formula at $n=5$, not its rate, and we record the observation without an asymptotic interpretation.

S3.3. *The direct evaluation at a negative saddlepoint.* In Kolassa and Li's example, at the two points with $\hat{s}_1<0$ the values attributed to Kolassa (2003) are not his approximation as he prescribes it: he requires, as in (17), that the component with a negative saddlepoint be negated and the probability obtained by differencing, which gives our order one. The printed values, 1.41×10^{-2} and 3.99×10^{-3}, are reproduced, as 1.4112×10^{-2} and 3.9871×10^{-3}, by the first-order formula evaluated directly at the negative saddlepoint. Section 3.3 of the paper explains why this evaluation has the wrong exponent; at $n=5$ the shortfall (22) of the paper is only $\Delta=1.9\times10^{-3}$ and 1.3×10^{-2}, which is why the direct values look reasonable there, and at $(2.5,3.5)$ even better than our order one. Table S6 follows the two forms as n increases with y fixed, so that $\hat{s}$ is fixed. The error of the differenced form changes sign once and then settles to halving with each doubling of n, as an $O(n^{-1})$ error eventually should; that of the direct form does not decrease, changes sign, and reaches -2.2% and -50% at $n=640$. Beyond the table, at $y=(2.5,4.0)$ the logarithm of the ratio of the direct value to the exact one is -4.8, -11.0 and -24.1 at $n=2560$, 5120 and 10240, against $-\Delta=-6.9$, -13.7 and -27.5; the difference grows only like $\log n$, so the logarithm of the ratio falls at the rate Δ/n per observation, as (22) predicts.

S3.4. *Near the singular lines of Kolassa and Li's formula.* Kolassa and Li (2010) omit the two points of their example at which $\hat{s}_1=0$ (Table 2 of the paper), where their approximation has a removable singularity. We implemented their bivariate approximation as written out by Li (2009, Chapter 3), with the coordinates in their order; it reproduces the eight printed values to the three digits given at five points and to within two units in the third digit at the

TABLE S5

Kolassa and Li's example with y fixed and n doubling: exact $\mathbb{P}(\bar{Y}_1 \geq y_1, \bar{Y}_2 \geq y_2)$ and relative errors of their approximation (KL, our implementation in the order of the coordinates of their paper; — on the line $\hat{s}_1 = 0$, where it is singular) and of the expansions of orders 1 to 5 (notation $a_k = a \times 10^k$). The saddlepoint is in the open orthant at the first two points and has $\hat{s}_1 = 0$ at the third. Entries below 10^{-12} in absolute value are at the accuracy of the exact values.

n	exact	KL	1	2	3	4	5
$y = (3.0, 3.0)$							
5	2.215196_{-2}	-7.6_{-3}	-3.3_{-2}	-3.8_{-4}	9.8_{-5}	-1.5_{-6}	-2.1_{-6}
10	4.552263_{-3}	-2.8_{-3}	-1.7_{-2}	-2.0_{-4}	1.4_{-5}	3.5_{-7}	-9.7_{-8}
20	2.539886_{-4}	-3.6_{-4}	-8.4_{-3}	-8.0_{-5}	1.7_{-6}	6.0_{-8}	-3.0_{-9}
40	1.147697_{-6}	9.2_{-4}	-3.9_{-3}	-2.8_{-5}	1.7_{-7}	6.5_{-9}	-5.1_{-11}
80	3.720233_{-11}	1.6_{-3}	-1.8_{-3}	-9.3_{-6}	6.0_{-9}	5.4_{-10}	1.5_{-12}
160	6.702771_{-20}	1.9_{-3}	-8.4_{-4}	-2.9_{-6}	-2.4_{-9}	3.2_{-11}	1.5_{-13}
$y = (3.5, 4.0)$							
5	1.655068_{-3}	9.1_{-3}	-2.7_{-2}	-6.3_{-4}	9.5_{-5}	6.6_{-6}	-2.8_{-6}
10	4.060597_{-5}	9.8_{-3}	-1.2_{-2}	-2.0_{-4}	1.2_{-5}	7.8_{-7}	-8.1_{-8}
20	3.532925_{-8}	9.7_{-3}	-5.6_{-3}	-6.1_{-5}	1.4_{-6}	6.9_{-8}	-2.0_{-9}
40	4.191728_{-14}	9.3_{-3}	-2.6_{-3}	-1.8_{-5}	1.8_{-7}	5.6_{-9}	-4.6_{-11}
80	9.910197_{-26}	9.0_{-3}	-1.2_{-3}	-5.5_{-6}	2.3_{-8}	4.4_{-10}	-9.2_{-13}
160	9.843574_{-49}	8.7_{-3}	-6.0_{-4}	-1.6_{-6}	3.0_{-9}	3.5_{-11}	-2.1_{-15}
$y = (2.5, 3.0)$							
5	4.123179_{-2}	—	-1.9_{-2}	-2.7_{-4}	4.2_{-5}	4.9_{-7}	-9.7_{-7}
10	1.264618_{-2}	—	-8.3_{-3}	-1.0_{-4}	4.4_{-6}	1.6_{-7}	-3.2_{-8}
20	1.437781_{-3}	—	-3.1_{-3}	-2.8_{-5}	3.5_{-7}	1.5_{-8}	-8.6_{-10}
40	2.385655_{-5}	—	-8.9_{-4}	-5.6_{-6}	2.7_{-8}	9.3_{-10}	-2.1_{-11}
80	8.837022_{-9}	—	-1.6_{-4}	-9.4_{-7}	2.9_{-9}	5.7_{-11}	-5.9_{-13}
160	1.682751_{-15}	—	3.3_{-5}	-1.5_{-7}	3.5_{-10}	3.6_{-12}	-2.8_{-14}

TABLE S6

Kolassa and Li's example at the two points with $\hat{s}_1 < 0$, with y fixed and n increasing: exact orthant probability and relative errors of the first-order formula applied directly at the negative saddlepoint (direct, as in the K03 column of Table 2 of the paper) and with the reduction (17) of the paper (reduced, our order one) (notation $a_k = a \times 10^k$). At $n = 5$ the direct errors differ slightly from the K03 column of Table 2, which is computed from the three-digit printed values.

	$y = (2.5, 3.5)$			$y = (2.5, 4.0)$		
n	exact	direct	reduced	exact	direct	reduced
5	1.4127_{-2}	-1.1_{-3}	-6.5_{-3}	3.9337_{-3}	1.4_{-2}	-8.3_{-4}
10	1.7240_{-3}	4.5_{-3}	-8.0_{-4}	1.5015_{-4}	1.4_{-2}	1.3_{-3}
20	3.0738_{-5}	5.5_{-3}	6.7_{-4}	2.5838_{-7}	1.2_{-2}	1.2_{-3}
40	1.2006_{-8}	5.0_{-3}	6.9_{-4}	9.2382_{-13}	7.1_{-3}	7.9_{-4}
80	2.2667_{-15}	4.1_{-3}	4.3_{-4}	1.4772_{-23}	-7.4_{-3}	4.5_{-4}
160	1.0044_{-28}	2.4_{-3}	2.3_{-4}	5.0219_{-45}	-5.6_{-2}	2.4_{-4}
320	2.5279_{-55}	-2.8_{-3}	1.2_{-4}	8.0750_{-88}	-2.0_{-1}	1.3_{-4}
640	2.1735_{-108}	-2.2_{-2}	6.3_{-5}	2.9380_{-173}	-5.0_{-1}	6.4_{-5}

other three (the second derivative of the nested signed root that the approximation requires is computed by central differences with step $10^{-3}\max(1, |\hat{s}_1|)$; steps three and ten times larger change the result by less than 10^{-7} relatively). The approximation contains, as (S1) with $M = 1$ does, differences of the reciprocals of a standardized saddlepoint component and of a signed root, for the first coordinate at $\hat{s}$ and for the second at the saddlepoint of the marginal of $\bar{Y}_2$, and these are $\infty - \infty$ on the lines $\hat{s}_1 = 0$ and $y_2 = E(\bar{Y}_2)$. The limits are finite, but Li (2009, Section 3.1) omits the formulas for these special cases as algebraically involved, and in double precision the formula loses all accuracy near the lines (Table S7):

along $y = (2.5 + \delta, 3.0)$, where $\hat{s}_1 \approx 2.35\delta$, its relative error is -0.46% for $\delta \geq 10^{-4}$, 6.9% at $\delta = 10^{-5}$, and grows by about three orders of magnitude per decade of δ below that, roughly as $\epsilon/\hat{s}_1^3$ with $\epsilon = 2.2 \times 10^{-16}$, because the signed root is the square root of a difference of values of K that is $O(\hat{s}_1^2)$; along $y = (2.5, 2.0 + \delta)$ the second difference behaves in the same way. The affected band is narrow, about 10^{-5} in y against standard deviations of 0.63, but it contains the points at which a coordinate is at its mean, and Kolassa and Li's two omitted points. The approximation itself has finite limits on the lines, and the loss comes from the evaluation of the formula in floating point; an expansion about the lines, which Li (2009) omits, would remove it, as the series for b_i in Section S1 does for (S1). Reversing the order of the coordinates avoids the singularity in each case, at the cost of the dependence on the order: at the four points of Table 2 of the paper with $y_1 \neq y_2$ the two orders differ by 0.9 to 1.1 percentage points, as much as the error itself, -0.01% against -0.97% at $(3.0, 3.5)$ and 0.91% against -0.22% at $(3.5, 4.0)$, and neither order is uniformly better. Our expansion involves no signed roots and no division by a component of $\hat{s}$, does not depend on the order of the coordinates, and its error is smooth along both sweeps of Table S7.

TABLE S7

Kolassa and Li's approximation near its singular lines, $n = 5$: relative errors of their approximation evaluated directly from its formula in double precision (KL) and of the expansion of order 2 of the paper along $y = (2.5 + \delta, 3.0)$, which approaches $\hat{s}_1 = 0$, and along $y = (2.5, 2.0 + \delta)$, which approaches $y_2 = E(\bar{Y}_2)$ (notation $a_k = a \times 10^k$).

	$y = (2.5 + \delta,\ 3.0)$			$y = (2.5,\ 2.0 + \delta)$	
δ	$\hat{s}_1$	KL	2	KL	2
10^{-2}	2.3_{-2}	-4.6_{-3}	-2.6_{-4}	-1.4_{-2}	-1.5_{-4}
10^{-3}	2.4_{-3}	-4.6_{-3}	-2.7_{-4}	-1.4_{-2}	-1.5_{-4}
10^{-4}	2.4_{-4}	-4.7_{-3}	-2.7_{-4}	-1.4_{-2}	-1.5_{-4}
10^{-5}	2.4_{-5}	6.9_{-2}	-2.7_{-4}	-7.6_{-2}	-1.5_{-4}
10^{-6}	2.4_{-6}	2.7_{2}	-2.7_{-4}	2.3_{1}	-1.5_{-4}
10^{-7}	2.4_{-7}	-4.9_{5}	-2.7_{-4}	5.2_{4}	-1.5_{-4}
10^{-8}	2.4_{-8}	$-\infty$	-2.7_{-4}	1.8_{7}	-1.5_{-4}
0	0	4.3_{16}	-2.7_{-4}	∞	-1.5_{-4}

S3.5. *The Laplace example.* Let $X = Z + A$ and $Y = Z + B$ as in Section 5.2 of the paper, with $K(s) = -\log\{1 - (s_1 + s_2)^2\} + (s_1^2 + s_2^2)/2$ and saddlepoint $\hat{s} = (x, -x)$ at the point $(x, -x)$, and consider the values of the approximations of Wang (1990) and Kolassa and Li (2010) quoted there (Table 3 of the paper). Two remarks on these values. First, the term (18) of Wang's Theorem 1 is written as a normal density at $\bar{v}_0$ with $\bar{v}_0^2 = -2\{K(0, \hat{s}_2) - \hat{s}_2 y\}$, where $K(0, \cdot)$ is the cumulant generating function of $\bar{Y}$ and $\hat{s}_2$ the second component of the joint saddlepoint. The function $u \mapsto uy - K(0, u)$ is concave, vanishes at $u = 0$ and is maximal at the saddlepoint of the marginal of $\bar{Y}$, so $\bar{v}_0^2$ would be nonnegative if $\hat{s}_2$ were that marginal saddlepoint, but the dependence between $\bar{X}$ and $\bar{Y}$ moves the joint saddlepoint away from it, here far enough that $\bar{v}_0^2 < 0$ at every point of Table 3 of the paper: $\bar{v}_0^2 = n\{x^2 + 2\log(1 - x^2)\}$, because the marginal saddlepoint of $\bar{Y}$ at $-x$ is about $-x/3$ while $\hat{s}_2 = -x$. The density is only a way of writing the factor $\exp[n\{K(0, \hat{s}_2) - \hat{s}_2 y\}]$ that his derivation produces (his equation (23)), and the column W evaluates the term through that factor, which exceeds one here. Second, at $x = 0.8$ the formula of Kolassa and Li (2010) is defined but no longer meaningful: its correction terms are as large as its leading term, which for $n = 10$ is by itself too small by a factor of thirty, and in the reversed order of the coordinates it returns a negative value. Both formulas evaluate K at a point that approaches the boundary of the strip as x increases to 1.

S4. Sample autocorrelations at two lags. This example, a pair of ratios of quadratic forms, follows the conventions of Section 5 of the paper; its quadratic forms grow in dimension with N, so it is a numerical application outside the setting of the theorems.

Let $z \sim N(0, \Sigma)$ with $\Sigma_{ts} = \rho^{|t-s|}$, $t, s = 1, \dots, N$, be N consecutive observations of a stationary first-order autoregression, and let $r_j = z^\top M A_j M z / z^\top M z$, $j = 1, 2$, be the mean-corrected sample autocorrelations at lags one and two, where $M = I - 11^\top/N$ with 1 a vector of ones, and $(A_j)_{ts} = 1/2$ for $|t - s| = j$ and 0 otherwise. The numerators and the common denominator are quadratic forms in z, so

$$\mathbb{K}(s,t) = -\tfrac{1}{2} \log |I - 2\Sigma^{1/2}\{s_1 M A_1 M + s_2 M A_2 M + (t_1 + t_2) M\}\Sigma^{1/2}|, \tag{S18}$$

with $\Sigma^{1/2}$ symmetric, as are all matrix square roots below; its derivatives follow from the recursion of Section 5.5 of the paper. Since $\mathbb{K}$ depends on t only through $u = t_1 + t_2$, the derivatives of $g_0 = \mathbb{K}^2_{\{1\}}$ and $g_1 = \mathbb{K}_{\{1,2\}}$ along $u = -c_1 s_1 - c_2 s_2$ follow by the chain rule. The first-order approximations are those of Butler and Paolella (1998). Exact values were obtained by two-dimensional numerical inversion of (9) of the paper and (45) of the paper on contours through the saddlepoint, by the trapezoidal rule, which converges exponentially for these analytic integrands; the double-precision values agree with a repetition in 34-digit arithmetic to 1.4×10^{-13} relative to F (3.6×10^{-13} relative to $\min(F, 1 - F)$). Table S8 reports $N = 50$ and 100 for white noise, $\rho = 0$, and for $\rho = 0.5$, at the points $c_j = \rho^j + q_j N^{-1/2}$.

The gains are smaller than in the univariate examples, but substantial overall for white noise, with occasional deterioration at the highest order: doubling N typically divides the error of order M by 2^M or more, and at $N = 100$ order five gives eight or nine digits for the distribution function and six to eight for the density. For $\rho = 0.5$, which reduces the effective sample size, the gains diminish after order two or three: at $N = 100$ the errors at order five are 4×10^{-8} to 2×10^{-5}, while at $N = 50$ the gains stop after order three, where the errors are 4×10^{-6} to 4×10^{-3}, and at the points $(1, 2)$ and $(2, 2)$ the errors of orders four and five are larger, up to 3×10^{-2}, as for the univariate distributions with few degrees of freedom. The expansions of Table S8 were computed in double precision in the whitened coordinates of Appendix C of the paper and Section S5: in the blocks with $\rho = 0.5$ the terms of the distribution function expansion in the original coordinates cancel by up to seventeen orders of magnitude, beyond the resolution of double precision, but by at most five in the whitened coordinates, and the double-precision results agree with a computation in 50-digit arithmetic to 1.3×10^{-12} relative to $\min(F, 1 - F)$ at every point of the table. The saddlepoint has a negative component at every point except $(q_1, q_2) = (-1, -1)$, so that (17) is used at all the other points, with no visible loss of accuracy.

S5. Computation: details.

S5.1. *Quadrature of the integrals $I(\nu)$.* In two dimensions the double-precision programs use the Gauss–Hermite representation of Appendix C of the paper, with 80×80 Gauss–Hermite nodes after a Cholesky transformation of W, when $\hat{s}_j \{(\Sigma^{-1})_{jj}\}^{-1/2} > 2.5$ for both j; the integrand is then analytic in a strip of at least that half-width in each coordinate, and the relative error is about 10^{-13} or less. Otherwise they evaluate $I(\nu)$ from its orthant-integral form

$$I(\nu) = \int_{t_1}^{\infty} \int_{t_2}^{\infty} \mathrm{He}_\nu(z - t, \Sigma)\, \phi(z, \Sigma)\, \mathrm{d}z_2\, \mathrm{d}z_1, \qquad \Sigma = n\hat{K}'', \quad t = \Sigma \hat{s}, \tag{S19}$$

by Gauss–Legendre quadrature with 200 nodes in each coordinate, in conditioned coordinates: the nodes in z_1 cover $[t_1, t_1 + 16\Sigma_{11}^{1/2}]$, and for each of them the nodes in z_2 cover the

TABLE S8

Sample autocorrelations at lags one and two of N observations of a first-order autoregression with parameter ρ: relative errors of the expansions of order M of the joint distribution function $\mathbb{P}(r_1 \le c_1, r_2 \le c_2)$ and of the density at $c_j = \rho^j + q_j N^{-1/2}$ (notation $a_k = a \times 10^k$). Asterisk: saddlepoint with at least one negative component, (17) used.

	distribution function, order M					density, order M				
(q_1, q_2)	1	2	3	4	5	1	2	3	4	5
$N = 50$, $\rho = 0$										
(-1,-1)	-3.7_{-2}	1.5_{-4}	1.4_{-5}	5.8_{-7}	-1.6_{-7}	-1.4_{-3}	7.3_{-5}	-1.0_{-5}	2.2_{-6}	-6.4_{-7}
(0.5,0.5)*	7.1_{-3}	-8.3_{-5}	-5.2_{-6}	1.1_{-6}	-3.7_{-7}	-1.9_{-3}	1.8_{-4}	-4.1_{-5}	1.5_{-5}	-6.9_{-6}
(1.5,-0.5)*	-4.6_{-3}	1.2_{-4}	-2.9_{-6}	1.1_{-7}	-2.5_{-8}	-1.7_{-3}	1.2_{-4}	-2.1_{-5}	6.1_{-6}	-2.4_{-6}
(-0.5,1.5)*	-3.5_{-3}	7.3_{-5}	3.3_{-6}	-1.9_{-6}	7.5_{-7}	-2.4_{-3}	2.9_{-4}	-8.2_{-5}	3.4_{-5}	-1.7_{-5}
(1,2)*	1.9_{-2}	-2.5_{-4}	2.0_{-6}	-7.0_{-7}	7.7_{-7}	-3.5_{-3}	6.0_{-4}	-2.0_{-4}	7.0_{-5}	-1.6_{-6}
(2,2)*	3.0_{-2}	-5.5_{-5}	-1.2_{-5}	-4.1_{-6}	5.3_{-6}	-3.8_{-3}	6.9_{-4}	-2.2_{-4}	5.7_{-5}	3.9_{-5}
$N = 100$, $\rho = 0$										
(-1,-1)	-1.9_{-2}	4.5_{-5}	2.4_{-6}	-2.0_{-8}	-3.9_{-9}	-3.7_{-4}	1.0_{-5}	-7.6_{-7}	8.5_{-8}	-1.3_{-8}
(0.5,0.5)*	3.2_{-3}	-2.1_{-5}	-3.4_{-7}	2.6_{-8}	-1.2_{-9}	-4.6_{-4}	2.0_{-5}	-2.1_{-6}	3.5_{-7}	-7.9_{-8}
(1.5,-0.5)*	-2.4_{-3}	3.2_{-5}	-3.7_{-7}	2.6_{-9}	-3.0_{-10}	-4.2_{-4}	1.5_{-5}	-1.4_{-6}	1.9_{-7}	-3.8_{-8}
(-0.5,1.5)*	-1.9_{-3}	2.1_{-5}	7.7_{-8}	-4.7_{-8}	7.8_{-9}	-5.3_{-4}	2.8_{-5}	-3.4_{-6}	6.5_{-7}	-1.7_{-7}
(1,2)*	8.6_{-3}	-5.5_{-5}	-1.2_{-7}	8.4_{-9}	-6.1_{-9}	-6.7_{-4}	4.6_{-5}	-6.9_{-6}	1.6_{-6}	-5.2_{-7}
(2,2)*	1.4_{-2}	-9.2_{-6}	-2.3_{-6}	5.1_{-8}	-6.8_{-9}	-7.2_{-4}	5.2_{-5}	-8.4_{-6}	2.1_{-6}	-7.2_{-7}
$N = 50$, $\rho = 0.5$										
(-1,-1)	-1.1_{-2}	6.8_{-4}	3.6_{-6}	-1.4_{-5}	5.6_{-6}	-2.3_{-3}	2.5_{-4}	-6.9_{-5}	2.8_{-5}	-1.3_{-5}
(0.5,0.5)*	2.9_{-2}	-1.9_{-3}	3.5_{-4}	-7.1_{-5}	-4.0_{-5}	-4.1_{-3}	7.2_{-4}	-2.2_{-4}	2.6_{-5}	9.0_{-5}
(1.5,-0.5)*	9.9_{-4}	1.8_{-5}	3.6_{-5}	-2.0_{-5}	6.7_{-6}	-4.4_{-3}	5.0_{-4}	1.1_{-5}	-3.0_{-4}	5.8_{-4}
(-0.5,1.5)*	-2.4_{-3}	7.7_{-5}	3.7_{-6}	-3.0_{-6}	-2.5_{-6}	-7.5_{-3}	1.2_{-3}	2.8_{-4}	-1.3_{-3}	4.8_{-3}
(1,2)*	4.5_{-2}	-1.1_{-3}	3.9_{-5}	4.7_{-4}	-1.4_{-3}	-8.7_{-3}	5.7_{-4}	1.9_{-3}	-6.9_{-3}	2.1_{-2}
(2,2)*	7.8_{-2}	4.6_{-3}	-4.0_{-3}	1.1_{-2}	-3.1_{-2}	-6.3_{-3}	1.1_{-3}	9.5_{-5}	-4.2_{-4}	6.7_{-4}
$N = 100$, $\rho = 0.5$										
(-1,-1)	-8.1_{-3}	1.9_{-4}	-3.2_{-7}	-4.3_{-7}	1.2_{-7}	-6.5_{-4}	4.0_{-5}	-6.0_{-6}	1.4_{-6}	-4.5_{-7}
(0.5,0.5)*	1.1_{-2}	-2.9_{-4}	2.3_{-5}	-4.7_{-6}	1.8_{-6}	-9.9_{-4}	9.2_{-5}	-1.9_{-5}	6.0_{-6}	-2.5_{-6}
(1.5,-0.5)*	-1.4_{-3}	6.1_{-5}	-2.1_{-7}	-4.8_{-7}	1.9_{-7}	-6.8_{-4}	3.3_{-5}	-3.7_{-6}	6.5_{-7}	-1.5_{-7}
(-0.5,1.5)*	-2.5_{-3}	5.1_{-5}	-1.3_{-6}	-3.6_{-8}	3.8_{-8}	-1.6_{-3}	2.3_{-4}	-6.3_{-5}	2.5_{-5}	-1.3_{-5}
(1,2)*	1.8_{-2}	-1.9_{-4}	1.5_{-5}	-6.6_{-6}	3.0_{-6}	-1.9_{-3}	2.9_{-4}	-8.8_{-5}	3.7_{-5}	-1.8_{-5}
(2,2)*	3.1_{-2}	1.1_{-4}	4.4_{-5}	-4.4_{-5}	2.2_{-5}	-1.5_{-3}	1.9_{-4}	-5.1_{-5}	2.0_{-5}	-9.7_{-6}

part above t_2 of an interval of 16 conditional standard deviations around the conditional mean of z_2 given z_1, which resolves the Gaussian ridge also when the components are strongly correlated. This avoids the binomial subtraction (14) of the paper; the Hermite polynomials change sign, so some cancellation remains in the quadrature sums, but it is mild at the points considered. When the conditional correlation is so close to ± 1 that the truncation $z_2 \ge t_2$ creates a boundary layer in z_1 narrower than a quarter of its range, the range of z_1 is split at the edges of the layer, with 200 nodes on each piece; for normal orthant probabilities at the origin this keeps the relative error below 10^{-9} for correlations up to $1 - 10^{-8}$ in absolute value, where a fixed set of nodes captures almost none of a small probability. Every retained term is compared with the same quadrature with half the nodes, and the number of nodes is doubled when the relative difference of the sum exceeds 10^{-8}; this comparison of two quadratures is not a bound, and gives no information on the asymptotic error.

In three dimensions the programs use the same two representations, with 48^3 Gauss–Hermite nodes when $\hat{s}_j\{(\Sigma^{-1})_{jj}\}^{-1/2} > 2.5$ for all j and otherwise 64 Gauss–Legendre nodes per coordinate in conditioned coordinates, the nodes in z_2 and z_3 covering the parts above t_2 and t_3 of intervals around the conditional means given the preceding coordinates, with the ranges of z_1 and z_2 split at boundary layers as in two dimensions, and they estimate the quadrature error by repeating the computation with half the nodes, doubling the number

of nodes once when the relative difference exceeds 10^{-8} (10^{-10} for Table 4 of the paper). Table 4 was computed in this way, in double precision, and checked at every point against the $I(\nu)$ computed from Lemma 2 and (14) in 40-digit arithmetic, with the bivariate and trivariate orthant probabilities of (15) obtained by Gauss–Legendre quadrature in conditioned coordinates with 80 nodes per dimension in the same precision; the two computations agree to the digits shown. At a point with $\hat{s}_j\{(\Sigma^{-1})_{jj}\}^{-1/2} = 3$ for all j, the 40-digit values of $I(\nu)$ with $|\nu| \le 15$ agree with the Gauss–Hermite representation with 60^3 nodes to 2×10^{-14} relative, and $I(0)$ with a standard trivariate normal routine to 3×10^{-15}. The cancellation in (12) grows faster with the order in three dimensions than in two, so the programs recompute the expansions in variable precision, with $\log_{10}\mathcal{C} + 26$ digits, whenever a double-precision indicator exceeds 10^6; this did not occur in Table 4. For $d \ge 4$ see Appendix C of the paper; at order five in three dimensions each of the 4060 integrals $I(\nu)$ is a sum over 64^3 or 128^3 nodes.

S5.2. *Whitening.* The whitened form (54) of the paper removes most of the cancellation among the terms of the expansion, as the indicators quoted in Appendix C show. Evaluated in the original coordinates on the same quadrature nodes, the order-five entries at the points $(3.5, 4.0)$ and $(4.0, 4.0)$ of Table 2 of the paper and the entries of orders four and five at the points $(1.5, 2.0)$ and $(2.0, 2.0)$ of Table 1 are lost to rounding (the values change by 3×10^{-6} and 2×10^{-5}, and by 10^{-9} to 3×10^{-7}, relative), while a 40-digit computation reproduces the whitened values to 5×10^{-14}. At a fixed point the cancellation in the original coordinates decreases as n grows, from 2.3×10^9 at $n = 5$ to 4.7×10^3 at $n = 160$ at the point $(3.5, 4.0)$ of Table S5.

Two details of the implementation affect the accuracy. The Taylor coefficients of K_Y should be computed directly rather than by substituting $L^{-\top}u$ into the polynomial in the original coordinates, which loses up to eight digits at degree eleven in the example: for quadratic forms K_Y is the cumulant generating function of the forms with matrices $\sum_{j=1}^{d}(L^{-\top})_{ji}B_j$, to which the trace recursion of Section 5.5 applies; and the Taylor coefficients of the amplitude of the ratio density of Section S4 are obtained from the same recursion and not from the chain rule, whose alternating sums lose five digits at $N = 50$. The indicator $\mathcal{C}$ concerns only the final sum and does not detect a loss of accuracy in the coefficients; when the coefficients are obtained by substitution, our programs estimate that loss separately, by propagating perturbations of the size of the rounding errors of the substitution through the expansion, and in three or more dimensions recompute the coefficients in variable precision when the estimate exceeds 10^{-10}. With these, the double-precision expansions agree with an independent computation in 50-digit arithmetic, in which all quantities (the matrices, the saddlepoints, the derivatives $\hat{K}_\nu$, the coefficients $c_{r,\nu}$, the $J(r)$ and $I(\nu)$ by Lemma 2 of the paper and (14), and the univariate terms of (17)) are evaluated to that precision, to 1.3×10^{-12} relative to $\min(F, 1 - F)$ for the distribution function and 10^{-13} for the density at every point of Table S8, and to 7×10^{-14} with a 40-digit computation at every point of Table 6.

S6. Programs. The zip archive accompanying this supplement contains the MATLAB programs that implement the expansions of the paper in one, two, three and more dimensions, the scripts and data that reproduce the tables of the paper and of this supplement, and a user manual that describes the programs, their arguments and their use. The programs are also available at https://raymond-kan.github.io/research.html.

REFERENCES

BUTLER, R. W. (2007). *Saddlepoint Approximations with Applications*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511619083

BUTLER, R. W. and PAOLELLA, M. S. (1998). Approximate distributions for the various serial correlograms. *Bernoulli* **4** 497–518. https://doi.org/10.2307/3318663

BUTLER, R. W. and PAOLELLA, M. S. (2002). Calculating the density and distribution function for the singly and doubly noncentral F. *Statist. Comput.* **12** 9–16. https://doi.org/10.1023/A:1013160019893

JENSEN, J. L. (1995). *Saddlepoint Approximations*. *Oxford Statistical Science Series* 16. Oxford: Oxford University Press. https://doi.org/10.1093/oso/9780198522959.001.0001

KOLASSA, J. E. (2003). Multivariate saddlepoint tail probability approximations. *Ann. Statist.* **31** 274–286. https://doi.org/10.1214/aos/1046294465

KOLASSA, J. E. and LI, J. (2010). Multivariate saddlepoint approximations in tail probability and conditional inference. *Bernoulli* **16** 1191–1207. https://doi.org/10.3150/09-BEJ237

LI, J. (2009). Saddle Point Approximation. PhD thesis, New Brunswick: Rutgers, The State University of New Jersey. https://doi.org/10.7282/T37D2VCF

LUGANNANI, R. and RICE, S. O. (1980). Saddle point approximation for the distribution of the sum of independent random variables. *Adv. in Appl. Probab.* **12** 475–490. https://doi.org/10.2307/1426607

ROUTLEDGE, R. and TSAO, M. (1997). On the relationship between two asymptotic expansions for the distribution of sample mean and its applications. *Ann. Statist.* **25** 2200–2209. https://doi.org/10.1214/aos/1069362394

WANG, S. (1990). Saddlepoint approximations for bivariate distributions. *J. Appl. Probab.* **27** 586–597. https://doi.org/10.2307/3214543

YANG, B. and KOLASSA, J. E. (2002). Saddlepoint approximation for the distribution function near the mean. *Ann. Inst. Statist. Math.* **54** 743–747. https://doi.org/10.1023/A:1022455116811